Reinforcement Learning

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***Abstract*—We develop mathematical summaries behind the reinforcement-learning step in modern language-model alignment, starting from Monte Carlo policy gradients through trust-region methods to Proximal Policy Optimisation (PPO) and Direct Preference Optimisation (DPO). After deriving the policy gradient theorem, we analyse the bias and variance of its Monte Carlo estimator and characterise when variance-reduction techniques preserve unbiasedness. Identifying the advantage function as the baseline, we examine its behaviour under value-function misspecification and its interpretation when rewards are learned from preference data. Trust-region methods follow from Kakade and Langford's performance difference lemma, yielding the TRPO surrogate, its second-order local approximation via the Fisher information matrix, and the asymmetric structure of the KL constraint. PPO comes forth as a first-order approximation that replaces the KL constraint with per-sample clipping; we extract the implicit assumptions in the clipped objective and construct a scenario where clipping systematically biases learning against rare, high-value actions. We close with the RLHF pipeline including the Bradley-Terry preference model, derive the closed-form optimum of the KL-regularised objective, and show how DPO exploits this closed form to convert preference learning into a supervised classification problem. Three threads recur throughout: the score-mean-zero identity as the algebraic engine of every gradient rearrangement, the performance difference lemma as the theoretical anchor of trust-region methods, and the progression from on-policy through quasi-off-policy to fully off-policy methods as a sample-efficiency-versus-adaptivity trade-off.**

I. Introduction

OpenAI's pipeline for instruction tuning does its reinforcement learning step using Proximal Policy Optimisation (PPO) [1], which is an algorithm motivated by the Trust Region Policy Optimisation framework [2] and installs modern Reinforcement Learning from Human Feedback (RLHF). Direct Preference Optimisation (DPO) [3] came later to show the RL step itself can be replaced by a supervised objective, done by exploiting the closed form of the KL-regularised optimal policy. This report goes through the mathematical processes that connect Monte Carlo policy gradients to PPO and then to DPO, justifying each modelling choice along the way.

Throughout, the environment is a discounted Markov decision process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with state s_t, action a_t, transition $P(s_{t+1}|s_t, a_t)$, reward $r_t = r(s_t, a_t)$ and discount $\gamma \in [0, 1)$. A stochastic policy $\pi_\theta(a|s)$ parameterised by $\theta \in \mathbb{R}^d$ induces a trajectory distribution

$$p_\theta(\tau) = p(s_0) \prod_{t\geq 0} P(s_{t+1}|s_t, a_t)\, \pi_\theta(a_t|s_t), \tag{1}$$

on $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots)$. We work with the expected discounted return

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\geq 0} \gamma^t r_t\right], \tag{2}$$

and the state-value, action-value and advantage functions are defined in the usual way:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_\pi\left[\sum_{k\geq 0} \gamma^k r_{t+k} \,\middle|\, s_t = s\right], \\ Q^\pi(s, a) &= \mathbb{E}_\pi\left[\sum_{k\geq 0} \gamma^k r_{t+k} \,\middle|\, s_t = s, a_t = a\right], \end{aligned} \tag{3}$$

$$A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s). \tag{4}$$

II. Monte Carlo Estimation of Policy Gradients

A. Expectations under a Gaussian policy

Consider a one-dimensional Gaussian policy parameterised by $\theta = (\mu, \sigma)$:

$$\pi_\theta(a|s) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a - \mu s)^2}{2\sigma^2}\right), \tag{5}$$

with log-density

$$\log \pi_\theta(a|s) = -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{(a - \mu s)^2}{2\sigma^2}. \tag{6}$$

(i) Expected action $\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[a]$. Change variables to $u = (a - \mu s)/\sigma$, so that $u \sim \mathcal{N}(0, 1)$. Then

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[a] = \mathbb{E}[\mu s + \sigma u] = \mu s, \tag{7}$$

since $\mathbb{E}[u] = 0$. Equation (7) confirms that the mean of (5) is μs.

(ii) Expected log-policy $\mathbb{E}[\log \pi_\theta(a|s)]$. Substituting (6) and using $\mathbb{E}[(a - \mu s)^2] = \sigma^2$:

$$\mathbb{E}[\log \pi_\theta(a|s)] = -\tfrac{1}{2}\log(2\pi e\sigma^2). \tag{8}$$

Quantity (8) is the negative differential entropy of $\pi_\theta(\cdot|s)$, depending on σ alone; the mean μs drops out under centring.

(iii) Expected score with respect to μ. Differentiating (6),

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \mu} = \frac{s(a - \mu s)}{\sigma^2}. \tag{9}$$

Taking expectations under the policy and using $\mathbb{E}[a - \mu s] = 0$,

$$\mathbb{E}_{a\sim\pi_\theta}\left[\frac{\partial \log \pi_\theta}{\partial \mu}\right] = 0. \tag{10}$$

(iv) Expected score with respect to σ. Differentiating again,

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \sigma} = -\frac{1}{\sigma} + \frac{(a - \mu s)^2}{\sigma^3}. \tag{11}$$

Using $\mathbb{E}[(a-\mu s)^2]=\sigma^2$,

$$\mathbb{E}_{a\sim\pi_\theta}\left[\frac{\partial \log \pi_\theta}{\partial \sigma}\right] = -\frac{1}{\sigma}+\frac{\sigma^2}{\sigma^3}=0. \tag{12}$$

Both score components disappear in expectation, also the result generalises. For any differentiable density π_θ whose support does not depend on θ, exchange of gradient and integral gives the score-mean-zero identity:

$$\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)] = \nabla_\theta \int \pi_\theta(a|s)\, da = 0. \tag{13}$$

Identity (13) is useful for many algebraic steps that follow: baseline subtraction, advantage centring, and the local geometry of the KL divergence, they all reduce to it.

B. Policy gradient theorem

Expanding (2) as an integral over trajectories,

$$J(\theta)=\int p_\theta(\tau)\, R(\tau)\, d\tau, \qquad R(\tau)=\sum_{t\geq 0}\gamma^t r_t, \tag{14}$$

intuitively, we wish to push ∇_θ inside the integral. Doing so requires the Leibniz rule, which holds under three conditions: (a) $\theta \mapsto p_\theta(\tau)$ is differentiable for almost every τ; (b) there is an integrable dominating function $g(\tau)\geq 0$ with $\|\nabla_\theta p_\theta(\tau)\, R(\tau)\| \leq g(\tau)$ in a neighbourhood of the point of interest; (c) the support of p_θ does not vary with θ. Condition (c) is automatic whenever $\pi_\theta(a|s)>0$. Condition (b) holds when rewards are bounded and $\gamma<1$, since then $|R(\tau)|\leq r_{\max}/(1-\gamma)$ and $g(\tau)=\|\nabla_\theta p_\theta(\tau)\|\cdot r_{\max}/(1-\gamma)$ is integrable.

Given the interchange, multiply and divide by p_θ to obtain $\nabla_\theta p_\theta = p_\theta \nabla_\theta \log p_\theta$, so that

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}[\nabla_\theta \log p_\theta(\tau)\, R(\tau)]\,. \tag{15}$$

From (1) only the policy factors carry θ, hence

$$\nabla_\theta \log p_\theta(\tau) = \sum_{t\geq 0}\nabla_\theta \log \pi_\theta(a_t|s_t). \tag{16}$$

Substituting (16) into (15) and tightening via causality (action a_t cannot influence rewards earned before t, and the remaining return-from-t has conditional mean $Q^\pi(s_t,a_t)$) recovers the canonical form

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)\, Q^\pi(s,a)]\,, \tag{17}$$

where $d^\pi(s)=(1-\gamma)\sum_t \gamma^t P(s_t=s\mid\pi)$ is the discounted state visitation. Equation (17) is the policy gradient theorem of Sutton et al. The motivation is, an expectation over an unknown environment has been turned into one over quantities we can sample. Only one object must be differentiated analytically, the score $\nabla_\theta \log \pi_\theta$; everything else is observable from rollouts.

C. The Monte Carlo estimator: sources of bias and variance

Given N trajectories of horizon T sampled under π_θ, the Monte Carlo estimator of (17) is

$$\hat{g}=\frac{1}{N}\sum_{n=1}^{N}\sum_{t=0}^{T-1}\nabla_\theta \log \pi_\theta(a_t^{(n)}|s_t^{(n)})\,\hat{R}_t^{(n)}, \tag{18}$$

with $\hat{R}_t^{(n)}=\sum_{k=t}^{T-1}\gamma^{k-t}r_k^{(n)}$ being the return from time t. Under these conditions the estimator is unbiased: trajectories are on policy, the horizon is infinite (or γ^T is negligible), and $\hat{R}_t$ is computed without function estimation. Neglecting any will introduce bias.

Bias sources fall into several families. Truncating the rollout at finite T drops the tail $\sum_{k\geq T}\gamma^k r_k$, with bias of order $\gamma^T r_{\max}/(1-\gamma)$. Sampling trajectories under a behaviour policy distinct from π_θ without importance correction introduces a sampling mismatch. Replacing $\hat{R}_t$ with a bootstrapped target $r_t+\gamma\hat{V}(s_{t+1})$ inherits the value-function error as systematic bias. Function approximation in any learned critic propagates approximation error into the gradient estimate.

Long horizons amplify $\mathrm{Var}(\hat{R}_t)$ through accumulated reward noise. Sharp policies push $\|\nabla_\theta \log \pi_\theta\|$ to large values at low-probability actions, therein heightening score factor (this shows why variance is a practical concern). Finite samples leave lots of state space uncovered. Even within a single trajectory, returns at different t are correlated through the shared dynamics, so the effective sample size falls short of NT. Consistency follows under bounded second moments, but the standard error of $\hat{g}$ scales as $\sqrt{T/N}$, and long horizons need many rollouts.

D. Variance reduction without bias

Any baseline $b(s)$ that depends on the state alone may be subtracted from the return:

$$\hat{g}_b=\frac{1}{N}\sum_{n}\sum_{t}\nabla_\theta \log \pi_\theta(a_t^{(n)}|s_t^{(n)})\left(\hat{R}_t^{(n)}-b(s_t^{(n)})\right). \tag{19}$$

Unbiasedness follows from (13), pulling $b(s)$ out of the inner integral,

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(a|s)\, b(s)] = b(s)\,\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)] = 0, \tag{20}$$

so $\mathbb{E}[\hat{g}_b]=\mathbb{E}[\hat{g}]$. Variance can be cut by an appropriate choice of b; in the limit of a perfect critic, $b(s)=V^\pi(s)$ gives the advantage form and the gradient becomes $\mathbb{E}[\nabla_\theta \log \pi_\theta \cdot A^\pi(s,a)]$.

Bias slips back in whenever the variance-reduction step is not a pure function of state. Action dependent baselines $b(s,a)$ break (20) unless an analytical correction (a control variate) restores the score identity. Bootstrapped advantage estimators such as the generalised advantage estimator

$$\hat{A}_t^{\mathrm{GAE}(\lambda)}=\sum_{l\geq 0}(\gamma\lambda)^l\,\delta_{t+l}, \tag{21}$$

with TD residual $\delta_t=r_t+\gamma\hat{V}(s_{t+1})-\hat{V}(s_t)$, introduce bias proportional to the critic error when $\lambda<1$, trading variance for bias along the λ axis. Self-normalised importance

sampling (used when rolling data over multiple gradient steps) is consistent yet biased at finite sample size. Reuse of data after θ has changed is off-policy; the corresponding $\hat{g}$ is biased without an importance ratio. Avoiding these bias sources requires keeping baselines to state only, choosing λ close to one when bias matters more than variance, and applying importance corrections whenever data crosses a parameter update.

III. Using the Advantage Function for Policy Learning

A. The advantage has zero mean under the policy

Definition (4) sets $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$. By construction the value function is the policy-mean of the action-value:

$$V^\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)}[Q^\pi(s,a)]. \tag{22}$$

Subtracting (22) from $V^\pi(s)$ gives the identity

$$\mathbb{E}_{a\sim\pi}[A^\pi(s,a)] = V^\pi(s) - V^\pi(s) = 0, \qquad \forall s \in \mathcal{S}. \tag{23}$$

There are some consequences however. Firstly, V^π qualifies as a valid baseline for the policy gradient: replacing $b(s)$ with $V^\pi(s)$ in (19) produces an unbiased estimator (the argument is exactly (20)), so the advantage form

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi}[\nabla_\theta \log \pi_\theta(a|s)\, A^\pi(s,a)] \tag{24}$$

has the same mean as (17). Next, among all state-only baselines, V^π achieves close to optimal variance reduction. A variance-minimising choice is the score-weighted value $b^*(s) = \mathbb{E}[\|\nabla \log \pi\|^2 Q^\pi]/\mathbb{E}[\|\nabla \log \pi\|^2]$, and V^π closely tracks it for most policies of practical interest. Lastly, centring in (23) makes the gradient direction reward-shift invariant: adding a constant or a state-only baseline to r leaves A^π, and thus (24), unchanged. Without it, policy updates would track absolute reward magnitudes, void of algorithmic content.

B. Misspecified value functions

Suppose the critic carries additive state-dependent error

$$\tilde{V}(s) = V^\pi(s) + \varepsilon(s), \tag{25}$$

producing an approximate advantage $\tilde{A}(s,a) := Q^\pi(s,a) - \tilde{V}(s) = A^\pi(s,a) - \varepsilon(s)$. Substituting into the policy-gradient expectation,

$$\mathbb{E}\Big[\nabla_\theta \log \pi_\theta\, \tilde{A}\Big] = \mathbb{E}[\nabla_\theta \log \pi_\theta\, A^\pi] - \mathbb{E}[\nabla_\theta \log \pi_\theta\, \varepsilon(s)]. \tag{26}$$

Score-mean-zero applied conditionally on s reduces the second term to zero, so the gradient direction does not change; a state-only critic error introduces no bias in ∇J.

Variance is different, and here we show why. Computing the variance of the right-hand side of (26),

$$\begin{aligned}\mathrm{Var}(\nabla \log \pi \cdot \tilde{A}) = \mathrm{Var}(\nabla \log \pi \cdot A^\pi) \\ - 2\,\mathrm{Cov}(\nabla \log \pi \cdot A^\pi,\, \nabla \log \pi \cdot \varepsilon) \\ + \mathrm{Var}(\nabla \log \pi \cdot \varepsilon).\end{aligned} \tag{27}$$

Non-negativity of the last term gives a variance that grows with $\|\varepsilon\|$, so a noisy critic costs samples without misleading the policy.

Robustness fails as soon as the error gains action-dependence. If $\tilde{V}$ becomes $\tilde{V}(s,a)$ or if Q^π itself is approximated, the score-trick argument no longer kills the cross-term, and the gradient picks up a bias of order $\mathbb{E}[\nabla \log \pi \cdot \delta(s,a)]$. Actor-critic methods accept this bias because the variance reduction from a learned critic outweighs it in typical regimes. Trading bias for variance through λ-returns (21) formalises the same idea: $\lambda = 1$ yields unbiased high-variance Monte Carlo, $\lambda = 0$ yields biased low-variance temporal-difference, with intermediate λ giving a tunable middle ground.

C. Interpretability under learned rewards

Under a learned reward $r_\phi(s,a)$, as in RLHF, the advantage (4) is computed with respect to value functions V^π_ϕ, Q^π_ϕ that use r_ϕ in place of the true reward r^*. Interpretability shifts in three ways.

A high value of A^π_ϕ no longer signals a human-preferred action; it signals an action the reward model believes to be good. Calibration of r_ϕ bounds the divergence between these two notions. A reward model trained on a finite preference dataset with annotator or length bias, or stylistic preferences will bias A^π_ϕ in those same directions.

Out-of-distribution evaluations are unreliable. Reward models are trained on samples from a particular policy distribution (typically the supervised fine-tuned model π_{SFT}), and their reward estimates extrapolate as the deployed policy drifts. Wherever the current policy visits regions r_ϕ has not seen, A^π_ϕ has arbitrary sign and magnitude. Reward hacking is the predictable outcome; the policy gradient pushes towards (s,a) pairs where A^π_ϕ is large but A^* is small or negative. KL regularisation against π_{SFT}, discussed in Section VI-A, is there to keep π inside the region where the equivalence between A^π_ϕ and A^* approximately holds.

Bayesian or ensemble reward models give calibrated uncertainty $\mathrm{std}(r_\phi)$ that can be made into an advantage $A^\pi_\phi - \kappa\,\mathrm{std}(r_\phi)$, penalising high-uncertainty actions. DPO sidesteps the advantage altogether, deriving its update from a preference likelihood (Section VI).

IV. Trust Regions and the KL Divergence

A. Why the KL appears in the TRPO objective

Starting from the performance difference lemma of Kakade and Langford: for any two policies π and $\tilde{\pi}$,

$$\begin{aligned}J(\tilde{\pi}) - J(\pi) &= \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t\geq 0}\gamma^t A^\pi(s_t,a_t)\right] \\ &= \tfrac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^{\tilde{\pi}},\, a\sim\tilde{\pi}}[A^\pi(s,a)].\end{aligned} \tag{28}$$

Equation (28) is exact, but we must sample from $d^{\tilde{\pi}}$ to evaluate its right-hand side, the visitation of the policy we wish to optimise. Such samples are not available before the update.

TRPO [2] replaces $d^{\tilde{\pi}}$ with d^π, defining the surrogate

$$L_\pi(\tilde{\pi}) := J(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi,\, a\sim\tilde{\pi}}[A^\pi(s,a)]. \tag{29}$$

Matching of J holds to first order around $\tilde{\pi} = \pi$: $L_\pi(\pi) = J(\pi)$ and $\nabla L_\pi(\pi) = \nabla J(\pi)$. Tractability is restored because L_π can be evaluated from samples under π via importance sampling. Accuracy of (29) as a proxy for $J(\tilde{\pi})$ holds only while $\tilde{\pi}$ stays close to π. Unrestrained optimisation of L_π exploits the approximation error and returns spurious gains.

Correction comes via the Kakade-Langford bound, sharpened in [2]:

$$J(\tilde{\pi}) \geq L_\pi(\tilde{\pi}) - C \max_s D_{\mathrm{KL}}(\pi(\cdot|s) \,\|\, \tilde{\pi}(\cdot|s)),$$
$$C = \frac{2\gamma\epsilon_A}{(1-\gamma)^2}, \quad \epsilon_A = \max_{s,a} |A^\pi(s,a)|. \tag{30}$$

Maximising L_π subject to a KL constraint thus maximises a guaranteed lower bound on J. In practice the $\max_s$ is replaced by an expectation under d^π for sample-based tractability:

$$\max_{\tilde{\pi}} L_\pi(\tilde{\pi})$$
$$\text{s.t. } \bar{D}_{\mathrm{KL}}(\pi, \tilde{\pi}) := \mathbb{E}_{s\sim d^\pi}[D_{\mathrm{KL}}(\pi(\cdot|s) \,\|\, \tilde{\pi}(\cdot|s))] \leq \delta. \tag{31}$$

What the KL adds is honesty about locality: only step as far as the bound is tight.

B. Local approximation via second-order Taylor expansion

Let $\Delta = \theta - \theta_{\text{old}}$ and treat both terms of (31) as functions of θ around θ_{old}. Two facts make a low-order expansion natural, the first is, $\bar{D}_{\mathrm{KL}}(\theta_{\text{old}}, \theta_{\text{old}}) = 0$. Secondly, the gradient also vanishes there:

$$\nabla_\theta \bar{D}_{\mathrm{KL}}(\theta)\big|_{\theta_{\text{old}}} = -\mathbb{E}_{s,a}[\nabla_\theta \log \pi_\theta(a|s)]\big|_{\theta_{\text{old}}} = 0, \tag{32}$$

by the score-mean-zero identity (13). So the Taylor expansion of $\bar{D}_{\mathrm{KL}}$ begins at the quadratic term:

$$\bar{D}_{\mathrm{KL}}(\pi_{\theta_{\text{old}}}, \pi_\theta) \approx \tfrac{1}{2}\Delta^\top F(\theta_{\text{old}})\Delta + O(\|\Delta\|^3), \tag{33}$$

where the Fisher information matrix

$$F(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta \, \nabla_\theta \log \pi_\theta^\top\right] \tag{34}$$

arises as the Hessian after a brief score-identity computation. Concurrently, L_π admits the linear approximation $L_\pi(\pi_\theta) \approx L_\pi(\pi_{\theta_{\text{old}}}) + g^\top\Delta$, with $g = \nabla_\theta J(\theta_{\text{old}})$. Matching orders, problem (31) becomes

$$\max_\Delta \; g^\top\Delta \quad \text{s.t.} \quad \tfrac{1}{2}\Delta^\top F\,\Delta \leq \delta. \tag{35}$$

Solving (35) by Lagrange multipliers gives $g = \lambda F\Delta$, hence $\Delta = F^{-1}g/\lambda$. Substituting back into the active constraint yields

$$\Delta^\star = \sqrt{\frac{2\delta}{g^\top F^{-1} g}}\, F^{-1} g. \tag{36}$$

Equation (36) is the natural gradient step with step size tied to the KL budget δ. TRPO follows this direction and then applies a backtracking line search to confirm that both the surrogate has improved and the actual KL stays inside δ, since the quadratic in (33) can underestimate the true KL once $\|\Delta\|$ grows.

"Local approximation" has a precise meaning here. Working with a linear L_π and quadratic $\bar{D}_{\mathrm{KL}}$ is faithful only inside the ellipsoid $\frac{1}{2}\Delta^\top F\Delta \leq \delta$; outside it both approximations degrade. Reparameterisation invariance follows as a bonus: $F^{-1}g$ does not depend on the choice of θ-coordinates, unlike the raw gradient g.

C. When small KL fails to bound performance degradation

Bound (30) guarantees that a small KL keeps the surrogate close to the objective, modulated by the constant $C = 2\gamma\epsilon_A/(1-\gamma)^2$. Three regimes weaken the practical content of this guarantee.

Constant C blows up when γ approaches one or when advantages have a big dynamic range. In sparse-reward environments where a small number of states carry the bulk of the return, ϵ_A becomes large, and the KL budget needed to bound performance change shrinks. Conservative settings such as $\delta \sim 10^{-2}$ may exceed this by orders of magnitude.

A pointwise example can be given, by considering a state $s^\star$ at which one action $a^\dagger$ has $A^\pi(s^\star, a^\dagger) = -M$ for large M, with all other actions having zero advantage. Suppose $\pi(a^\dagger|s^\star) = 0.5$ and a candidate $\tilde{\pi}$ shifts this to 0.55. A quadratic approximation to the KL at this state gives

$$D_{\mathrm{KL}}(\pi\|\tilde{\pi})(s^\star) \approx \frac{(0.05)^2}{2 \cdot 0.5 \cdot 0.5} = 0.005, \tag{37}$$

a negligible number. Change in expected reward at $s^\star$ equals $0.05 \cdot (-M)$, which can be made arbitrarily bad by choice of M. So small KL coexists with large performance change whenever the advantage function carries narrow, deep cliffs.

A second failure mode comes from coverage. Expectation in $\bar{D}_{\mathrm{KL}}$ is taken under d^π, the current visitation. States that π rarely visits contribute little to this expectation, even if the new policy makes drastic changes there. Should $\tilde{\pi}$ visit those states frequently, the bound misses the relevant deviation entirely. This distributional-shift problem is intrinsic to on policy KL constraints. A third concern follows from the asymmetry of the KL: $D_{\mathrm{KL}}(\pi\|\tilde{\pi})$ penalises $\tilde{\pi}$ for putting low mass where π is large, but tolerates $\tilde{\pi}$ moving mass into action regions π had very low probability on; such a move can produce a small KL with large behavioural change.

D. Asymmetry of the trust region

TRPO uses the forward direction $\bar{D}_{\mathrm{KL}}(\pi_{\text{old}}\|\pi_{\text{new}})$. One of the reasons is sample availability: $\bar{D}_{\mathrm{KL}}(\pi_{\text{old}}\|\pi_{\text{new}}) = \mathbb{E}_{a\sim\pi_{\text{old}}}[\log \pi_{\text{old}} - \log \pi_{\text{new}}]$ is estimated from the data we have while the reverse KL would require samples from the policy we are about to construct. Geometric clarity is another reason, the Taylor expansion in (33) produces the Fisher matrix of π_{old} with a clean sign, and the gradient interpretation follows immediately. Then, conservativeness: the forward KL is mode-covering, heavily penalising π_{new} for assigning low mass to actions π_{old} considered likely. This direction preserves the exploration the current policy carries; the reverse KL would let the policy collapse onto a few high-advantage actions, the same weaknesses trust regions guard against.

Two practical consequences deserve attention. Firstly, the forward KL is finite at any a with $\pi_{\text{old}}(a|s) = 0$ regardless of

π_{new} at that point, so TRPO does not penalise the opening of new action support; restricting the closing of existing support only. Secondly, the asymmetry is inherited by PPO at the level of per-sample clipping (Section V): updates that would increase the probability of a high-advantage action are clipped, but only in the direction the advantage drives them.

V. From Trust Regions to Proximal Policies

A. Deriving the PPO clipped objective

Importance sampling rewrites the TRPO surrogate (29) as an expectation under π_{old}. With the per-sample probability ratio

$$r_t(\theta) := \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}, \tag{38}$$

the surrogate takes the form

$$L^{\text{IS}}(\theta) = \mathbb{E}_{s\sim d^{\pi_{\text{old}}},\, a\sim\pi_{\text{old}}}\left[r_t(\theta)\,\hat{A}_t\right], \tag{39}$$

where $\hat{A}_t$ is an advantage estimate computed under π_{old}, typically via the GAE expression (21). At $\theta = \theta_{\text{old}}$ the ratio equals one and (39) reduces to $\mathbb{E}[\hat{A}_t]$. Unconstrained maximisation of (39) drives r_t to infinity when $\hat{A}_t > 0$ and to zero when $\hat{A}_t < 0$, exploiting an approximation that breaks down beyond a small neighbourhood of π_{old}.

TRPO controls this drift with the KL constraint (31). PPO [1] replaces the explicit constraint with a clipping operation, producing the clipped objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\; \text{clip}(r_t(\theta),\,1-\varepsilon,\,1+\varepsilon)\hat{A}_t\Big)\Big], \tag{40}$$

with $\text{clip}(x, a, b) = \max(a, \min(b, x))$ and clipping width ε (commonly 0.2). Taking the minimum makes (40) a pessimistic surrogate, bounded above by (39); PPO optimises this lower envelope.

Behaviour breaks into four cases according to the sign of $\hat{A}_t$ and the position of r_t in $[1-\varepsilon, 1+\varepsilon]$. When $\hat{A}_t > 0$ and $r_t \le 1+\varepsilon$, the clipped argument coincides with r_t, the minimum returns $r_t\hat{A}_t$, and the gradient points in the natural direction of raising $\pi_\theta(a_t|s_t)$. When $\hat{A}_t > 0$ and $r_t > 1+\varepsilon$, the clipped term equals $(1+\varepsilon)\hat{A}_t$, the min selects this constant in θ, and the gradient vanishes; further increases yield no objective gain. Symmetric reasoning handles $\hat{A}_t < 0$ with the clip biting at $r_t = 1-\varepsilon$: pushing r_t below $1-\varepsilon$ stops contributing to the objective.

Connection to TRPO is informal but instructive. A KL-penalised version of (39) adds $-\lambda\,\bar{D}_{\text{KL}}$, and near θ_{old} the KL behaves quadratically in $r_t - 1$ since $\log r_t \approx r_t - 1$. PPO swaps the smooth quadratic penalty for a piecewise-linear ceiling and floor: outside the clip range, contributions are constant in θ. The substitution is heuristic, with no exact equivalence between the two regularisers, yet it preserves the qualitative property of restricting movement away from π_{old} while removing the second-order linear algebra TRPO requires.

B. Implicit assumptions about the policy update distribution

PPO's appeal lies in its compactness, but compactness hides some assumptions about the regime in which the algorithm operates.

Modest distribution shift is implicit in the importance-sampling estimator (39): variance of r_t blows up when the new and old policies disagree, so the clip range ε must keep both close on the support of the data. A second assumption concerns advantage estimates: $\hat{A}_t$ is computed under π_{old}, and reusing this $\hat{A}_t$ across multiple gradient steps presumes that the advantage does not shift much across the clip range. Independent sampling is assumed at the level of the expectation in (40); within trajectories, returns at different t remain correlated through dynamics, which biases naive variance estimates without affecting the gradient itself.

A fourth assumption ties ε to a KL constraint: $|r_t - 1| \lesssim \varepsilon$ corresponds to a tolerable expected KL only for typical neural-network policies on standard benchmarks. Sharp policies or high-dimensional action spaces can break the correspondence. Fifth, the minimum in (40) encodes pessimism, which is desirable in expectation but can stall updates whenever $\hat{A}_t$ underestimates the magnitude of the true advantage. Lastly, the autograd implementation must treat the clip as a stop-gradient outside the unclipped region; the analytical update assumes gradient flow respects the clip boundaries.

C. A scenario in which clipping biases learning

Consider a contextual bandit with a single state and two actions, $\mathcal{A} = \{a_1, a_2\}$, with rewards $r(a_1) = 0$ and $r(a_2) = 10$. An initial policy assigns $\pi_{\theta_0}(a_1) = 0.99$, $\pi_{\theta_0}(a_2) = 0.01$; the optimum places all mass on a_2, demanding a hundredfold increase in the probability of the second action. Set $\varepsilon = 0.2$.

Trajectories that sample a_2 (probability 0.01 per draw) carry positive advantage of approximately $\hat{A} \approx 9.9$, and the surrogate (40) would push r_t upward. Clipping caps the per-sample contribution at $r_t = 1.2$, allowing $\pi(a_2)$ to grow by at most a factor of 1.2 per update under this single sample's pressure, from 0.01 to 0.012. Samples that draw a_1 (probability 0.99) carry $\hat{A} \approx -10$ and pull $\pi(a_1)$ downward, with the clip biting at $r_t = 0.8$.

Asymmetry sits in the ratio scale. Reducing $\pi(a_1)$ from 0.99 to $0.99 \times 0.8 = 0.792$ is a substantial absolute move because $\pi(a_1)$ has mass to spare. Increasing $\pi(a_2)$ from 0.01 to 0.012 is a tiny absolute move because there was so little mass to begin with. Multiplicative caps suffice in symmetric regimes but break when the optimal policy demands concentration on initially rare actions. Across iterations, PPO learns slowly in this setting, and the iterates can stall at suboptimal policies that put more weight on a_2 than initial but less than optimal.

Three mitigations are available. Larger ε allows faster up-weighting at the cost of greater off-policy variance. Replacing the clip with an adaptive KL penalty (the "PPO-penalty" variant) removes the hard cap and allows larger updates on rare-but-strong evidence. Exploration bonuses raise the sampling rate of a_2 so the bias is overcome by sheer quantity of rewarded samples. None of these alters the structural fact that PPO

penalises rapid amplification of low-probability high-value actions; in sparse-reward or hard-exploration regimes, the bias is real and not always softened by hyperparameter tuning.

VI. Reward Modelling in Language Models and Off-Policy Learning

A. RLHF: modelling assumptions and optimisation steps

Standard RLHF pipelines run in three stages. A pre-trained language model is fine-tuned on demonstration data to produce π_{SFT}. Human comparisons are collected on responses sampled from π_{SFT}, yielding a dataset $\mathcal{D} = \{(x_i, y_i^w, y_i^l)\}$ of prompt-with-winner-loser triples, on which a parameterised reward model $r_\phi(x, y)$ is trained. Finally, the policy is optimised with PPO under a reward-plus-KL objective anchored to π_{SFT}.

Reward-model training uses the Bradley-Terry preference likelihood

$$P(y^w \succ y^l \mid x) = \sigma\big(r_\phi(x, y^w) - r_\phi(x, y^l)\big), \tag{41}$$

with sigmoid $\sigma(z) = 1/(1 + e^{-z})$, maximising the log-likelihood of preferences in $\mathcal{D}$. Identifiability holds only up to an additive function of x, which is harmless because policy optimisation depends on reward differences alone. At the RL stage we solve

$$\begin{aligned}\max_\pi \mathbb{E}_{x\sim\mathcal{D}_x,\, y\sim\pi(\cdot|x)}[r_\phi(x, y)] \\ - \beta\, \mathbb{E}_x[D_{\text{KL}}(\pi(\cdot|x) \,\|\, \pi_{\text{ref}}(\cdot|x))],\end{aligned} \tag{42}$$

where $\pi_{\text{ref}} = \pi_{\text{SFT}}$ and $\beta > 0$ tunes the regulariser strength.

Bradley-Terry imposes a logistic preference noise model and presumes that pairwise preferences are explained by a scalar reward; intransitive or context-dependent preferences violate this. A single reward function is assumed to suffice for alignment; multi-objective trade-offs (helpfulness against harmlessness, for instance) are folded into annotator judgement implicitly. Reference policy trust is implicit: wherever r_ϕ is uninformative, π defaults to π_{SFT} under the KL anchor.

Optimisation interplay is the core practical issue. Training of r_ϕ uses samples from π_{SFT}, while evaluation runs on outputs of a drifting π. As the RL step pushes π off-distribution, r_ϕ extrapolates, opening reward-hacking pathways: (x, y) pairs with high r_ϕ but low true preference. KL regularisation in (42) serves as the operational defence: it anchors π to the region where r_ϕ was trained and where its outputs can be trusted. Tuning β is the principal lever in deployment; too small leaves the policy free to hack, too large blocks any meaningful adaptation. SFT versus RL also pulls in conflicting directions, with imitation pushing towards the demonstration distribution and reward maximisation pushing towards whatever the reward model favours; the KL term mediates between the two.

Equation (42) admits a closed-form maximiser. Treating $\pi(\cdot|x)$ as a probability distribution and applying Lagrange multipliers with the normalisation constraint,

$$\pi^\star(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{r_\phi(x, y)}{\beta}\right), \tag{43}$$

where the partition function $Z(x) = \sum_y \pi_{\text{ref}}(y|x) \exp(r_\phi(x, y)/\beta)$ normalises the distribution. Partition function $Z(x)$ is intractable over language outputs, which is why RLHF resorts to PPO as an iterative approximation. Equation (43) also seeds the DPO construction in the next subsection.

A useful reading of (43) comes from maximum-entropy reinforcement learning. The KL-regularised objective (42) can be rewritten, up to a θ-independent constant, as an entropy-regularised expected reward with π_{ref} playing the role of a base measure. When π_{ref} is uniform on a finite action space, the optimum drops to the Boltzmann policy $\pi^\star \propto \exp(r_\phi/\beta)$, the soft-max policy of maximum-entropy RL and soft Q-learning. For general π_{ref}, $\pi^\star$ behaves like a Bayesian-style update: the reference acts as a prior and $\exp(r_\phi/\beta)$ as the likelihood, with β controlling how aggressively new evidence overrides the prior. Small β amplifies the reward signal and lets $\pi^\star$ depart from π_{ref}; large β keeps $\pi^\star$ close to the prior regardless of reward magnitude. This is the same temperature parameter that appears throughout soft RL, and explains why RLHF's β controls policy entropy at the optimum in a way that mirrors the temperature in Boltzmann exploration.

For deployment, three consequences follow. Reward models embed annotator and dataset biases (length, formality, sycophancy), and the deployed policy inherits them at amplified intensity. Removing the KL anchor produces models that score highly on r_ϕ but degrade in human judgement, a now-routine empirical finding. Iterative pipelines that retrain r_ϕ on outputs of the updated π can amplify drift unless preference data is refreshed at each iteration to track the moving distribution.

B. DPO: trust regions, off-policy optimisation, stability, bias

Rafailov et al. [3] observed that (43) can be inverted to express the reward in terms of policy log-ratios:

$$r_\phi(x, y) = \beta \log \frac{\pi^\star(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x). \tag{44}$$

Second term depends only on x. In any reward difference $r_\phi(x, y^w) - r_\phi(x, y^l)$, the partition function cancels, leaving a quantity entirely in terms of policy-to-reference ratios. Substituting this difference into the Bradley-Terry likelihood (41) gives a preference likelihood expressed directly in $\pi^\star$, with no reward model to train and no partition function to estimate.

DPO's loss takes the negative log-likelihood of preferences under this parameterisation, with $\pi^\star$ replaced by a learnable π_θ:

$$\begin{aligned}\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{\mathcal{D}}\Bigg[\log \sigma\Bigg(\beta \log \frac{\pi_\theta(y^w|x)}{\pi_{\text{ref}}(y^w|x)} \\ - \beta \log \frac{\pi_\theta(y^l|x)}{\pi_{\text{ref}}(y^l|x)}\Bigg)\Bigg],\end{aligned} \tag{45}$$

with the expectation taken over preference triples $(x, y^w, y^l) \sim \mathcal{D}$. Minimising (45) trains π_θ to be the KL-regularised optimum (43), with the reward implicit in the log-ratio against π_{ref}.

Trust-region structure enters through π_{ref}, exactly as it does through π_{SFT} in RLHF. Initialising $\pi_\theta = \pi_{\text{ref}}$ makes all log-ratios zero and the sigmoid argument zero, so $\mathcal{L}_{\text{DPO}} = \log 2$

uniformly; the gradient pushes $\log(\pi_\theta(y^w)/\pi_{\text{ref}}(y^w))$ up and $\log(\pi_\theta(y^l)/\pi_{\text{ref}}(y^l))$ down. Coefficient β controls the scale at which preference signals translate into log-ratio movement, mirroring its role in (42). Where TRPO and PPO enforce trust regions through constraints or clipping on importance ratios, DPO embeds the trust region in the loss itself: log-ratio movement away from π_{ref} is what the objective rewards.

Off-policy operation is intrinsic. Preferences in $\mathcal{D}$ are sampled under whatever distribution generated the candidate responses, typically π_{SFT}; no rollouts from π_θ enter the training loop. Implicit importance correction lives in the $\log(\pi_\theta/\pi_{\text{ref}})$ form, with no need for explicit importance weights on the dataset. Optimisation is reduced to supervised classification, with the variance and engineering complexity of policy-gradient learning sidestepped.

Stability gains from DPO compared to PPO-RLHF can be traced to four structural removals. No separate reward model means no distribution-shift problem between r_ϕ and π. A supervised loss has dramatically lower gradient variance than policy-gradient targets that ride on noisy returns. Absence of rollouts removes the exploration noise that PPO must tolerate. Absence of clipping removes the asymmetric bias of Section V-C. Hyperparameter complexity shrinks to little beyond β and the optimiser.

Stability comes paired with biases of its own. Six are worth listing. Bradley-Terry survives as the underlying preference model, so intransitive or non-logistic preferences remain misrepresented. Implicit reward $\beta \log(\pi_\theta/\pi_{\text{ref}})$ is constrained to log-ratios of policies in the parametric family, restricting the class of rewards DPO can express; an RLHF reward model, being a separate function, has more flexibility. Sensitivity to π_{ref} deepens: in RLHF the reference acts as a regulariser, but in DPO it also defines the implicit reward, so flaws in π_{SFT} propagate into the learned policy with no error-correction loop. Off-policy data distribution sets the prompt regime; DPO has no mechanism for collecting new comparisons under the updated policy unless the pipeline is iterated externally. Lack of exploration means behaviours absent from $\mathcal{D}$ remain absent from π_θ; DPO can only re-weight what the dataset already contains. Likelihood displacement, documented after the original DPO paper, captures a counter-intuitive failure mode: minimising (45) can lower the absolute log-likelihood of preferred responses while raising the relative log-ratio. An objective that penalises high π_{ref}-relative probability on dispreferred responses can be satisfied by depressing π_θ on the preferred response provided π_θ on the dispreferred response drops faster.

Bias and stability sit on opposite ends of a deliberate trade. DPO removes the iterative, on-policy adaptation of PPO-RLHF, gaining sample efficiency and gradient quality at the cost of dependence on a fixed dataset and a fixed reference policy. PPO-RLHF retains adaptivity through rollouts and reward shaping at the cost of harder training and more failure modes. Practical deployments increasingly mix the two, using DPO as an initialisation followed by PPO-RLHF refinement when iterative on-policy collection is feasible.

A direct response to the likelihood-displacement issue is Identity Preference Optimisation (IPO) of Azar et al. [5], which replaces the log-sigmoid in (45) with a squared loss on the implicit reward difference:

$$\mathcal{L}_{\text{IPO}}(\theta) = \mathbb{E}_{\mathcal{D}}\left[\left(\log \frac{\pi_\theta(y^w|x)}{\pi_{\text{ref}}(y^w|x)} - \log \frac{\pi_\theta(y^l|x)}{\pi_{\text{ref}}(y^l|x)} - \frac{1}{2\beta}\right)^2\right]. \tag{46}$$

A finite minimum at the target log-ratio difference $1/(2\beta)$ caps how far the optimum can drive the policy away from π_{ref}, preventing the unbounded growth in implicit-reward magnitude that drives DPO into likelihood displacement. Empirical work since 2024 finds IPO is less prone to mode collapse on small preference datasets, with the trade being slower convergence on cleanly separated comparisons where DPO's sharper sigmoid loss has more gradient signal to exploit.

VII. Bayesian Reward Models and Active Preference Learning

The pipelines analysed up to this point treat r_ϕ as a single deterministic estimate. Replacing the point estimate with a posterior $p(r \mid \mathcal{D})$ introduces calibrated epistemic uncertainty, supports risk-sensitive policy learning, and opens the door to actively choosing which preference queries to acquire next. These extensions tie RLHF back to the wider Bayesian and active-learning toolkit.

A. Bayesian reward models

A Bayesian reward model maintains a posterior over reward functions given preference data. Tractable approximations include ensembles of independently trained reward heads on bootstrapped data, Monte Carlo dropout, and variational neural networks. Each choice trades expressivity for compute.

Given the posterior, two summaries characterise the model at a candidate query (x, y):

$$\bar{r}(x, y) = \mathbb{E}_{r\sim p(\cdot|\mathcal{D})}[r(x, y)], \tag{47}$$

$$\sigma_r(x, y) = \sqrt{\text{Var}_{r\sim p(\cdot|\mathcal{D})}[r(x, y)]}. \tag{48}$$

Posterior standard deviation σ_r rises in regions of input space where preference data is sparse or contradictory, and falls where the preference signal is consistent.

Conservative policy learning uses (48) to penalise actions the model is unsure about. One construction is the lower-confidence-bound advantage,

$$\hat{A}^\pi_{\text{cons}}(s, a) = \hat{A}^\pi_{\bar{r}}(s, a) - \kappa\, \sigma_r(s, a), \tag{49}$$

with $\kappa > 0$ a tunable risk parameter and $\hat{A}^\pi_{\bar{r}}$ the advantage computed under the posterior-mean reward. Substituting (49) for A^π in the policy gradient (24) produces an update that systematically avoids high-uncertainty state-action pairs. Reward hacking from Section III-C corresponds precisely to a policy exploiting regions where σ_r is large; expression (49) closes this avenue at the cost of slower learning in regions where the posterior remains diffuse.

A Bayesian reformulation also clarifies the role of the KL anchor. Without uncertainty quantification, the KL penalty against π_{ref} is the only mechanism preventing distribution shift into untrusted regions. Modelling σ_r explicitly lets β be tuned more aggressively because $\kappa\sigma_r$ takes over some of the work the KL was previously doing alone.

B. Active preference acquisition

Preference data is expensive when comparisons require human evaluation of long responses. With a Bayesian reward model, active learning asks which next prompt-pair (x, y_1, y_2) to query so as to maximally reduce posterior uncertainty.

The principled criterion is the expected information gain about r from observing the preference outcome c,

$$\mathrm{IG}(x, y_1, y_2) = H[r \mid \mathcal{D}] - \mathbb{E}_c[H[r \mid \mathcal{D} \cup \{c\}]], \tag{50}$$

where $c \in \{y_1 \succ y_2,\ y_2 \succ y_1\}$ and $H[\cdot]$ denotes entropy. Direct evaluation of (50) is rarely feasible, so the BALD criterion of Houlsby et al. rewrites it as

$$\mathrm{IG}(x, y_1, y_2) = H[c \mid \mathcal{D}] - \mathbb{E}_{r\sim p(\cdot\mid\mathcal{D})}[H[c \mid r]], \tag{51}$$

i.e., the predictive entropy of the preference minus the expected entropy under each posterior sample. Equation (51) is large precisely when the marginal preference is uncertain *and* individual posterior samples are confident on disagreeing answers: queries on which the ensemble would disagree.

Implementation collapses neatly when the posterior is represented as an ensemble of M reward models. Sample M preference probabilities $\sigma(r^{(m)}(x, y_1) - r^{(m)}(x, y_2))$, compute the marginal mean and the per-sample entropies, and take their difference. Selecting queries by (51) concentrates labelling budget on responses near the reward-model decision boundary, where each new preference label maximally reshapes the posterior.

Thompson sampling offers a related strategy with weaker theoretical guarantees but stronger empirical performance in non-stationary regimes: draw $r^{(m)} \sim p(r \mid \mathcal{D})$ at each step and query the pair this sampled reward most prefers. The connection between active acquisition and posterior sampling carries through to the policy: an analogous Thompson-sampling-flavoured policy gradient samples a reward from the posterior at each iteration and runs PPO against it, propagating epistemic uncertainty from reward model to policy without the conservatism of (49).

For RLHF specifically, active acquisition shifts the bottleneck from annotation budget to model training. Pipelines combining an ensemble reward model with BALD-style query selection can reduce the number of human comparisons needed to reach a fixed accuracy target by an order of magnitude, conditional on consistent annotators and a faithful posterior approximation.

VIII. Discussion

Here we discuss three threads throughout the development of this report. Identity (13) underpins every result that involves rearranging expectations of policy gradients, granting the freedom to subtract baselines without bias, making the advantage (23) the choice for variance-reducing, and forcing the Taylor expansion of the KL (33) to begin at second order, producing the Fisher matrix as a direct consequence. Identifying this single algebraic fact as a recurring factor clarifies why methods as distinct as REINFORCE, actor-critic, TRPO, PPO and DPO share a common core.

Performance difference lemma (28) is the second thread. Trust-region methods exist because (28) is exact but intractable, the surrogate (29) is tractable but approximate, and the divergence between them shrinks as policies stay close. TRPO enforces closeness with a KL constraint, PPO heuristically replaces this with per-sample clipping, and DPO exploits the closed-form optimum (43) of the KL-regularised problem to skip iterative optimisation entirely. Each method answers the same question with a different instrument.

A third thread tracks the move from on-policy to off-policy as we travel from Section II to Section VI. Monte Carlo policy gradients require fresh rollouts. PPO permits limited reuse through importance sampling, with the clip stopping the ratio from drifting too far. DPO is fully off-policy: training data is fixed at collection time and never updated by the policy. The bias-variance trade-off shifts accordingly, with each step gaining sample efficiency and losing adaptivity.

Open questions on the LLM side stem from the same trade-off. Iterative DPO variants, online reward model refinement, and on-policy preference collection all sit on the spectrum between full off-policy DPO and full on-policy RLHF. Conservatism in the form of π_{ref} remains the central tool against reward hacking; the question of how to relax this conservatism without losing the alignment guarantees the KL anchor provides is open.

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Appendix

Start from (2) and (3), which together give $J(\pi) = \mathbb{E}_{s_0\sim p_0}[V^\pi(s_0)]$. Since s_0 is drawn from the same initial

distribution under π and $\tilde{\pi}$,

$$J(\tilde{\pi}) - J(\pi) = \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t\geq 0}\gamma^t r_t - V^\pi(s_0)\right]. \tag{52}$$

Apply the telescoping identity

$$-V^\pi(s_0) = \sum_{t\geq 0}\left[\gamma^{t+1}V^\pi(s_{t+1}) - \gamma^t V^\pi(s_t)\right], \tag{53}$$

valid whenever $\gamma^T V^\pi(s_T) \to 0$ as $T \to \infty$, which holds under bounded rewards and $\gamma < 1$. Substituting (53) into (52) and grouping by power of γ,

$$J(\tilde{\pi}) - J(\pi) = \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t\geq 0}\gamma^t\big(r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t)\big)\right]. \tag{54}$$

The bracketed quantity is the temporal-difference residual against V^π. Conditioning on (s_t, a_t) and using the Bellman expectation identity, $\mathbb{E}_{\tilde{\pi}}[r_t + \gamma V^\pi(s_{t+1}) \mid s_t, a_t] = Q^\pi(s_t, a_t)$, the TD residual collapses to $A^\pi(s_t, a_t)$, yielding the trajectory form

$$J(\tilde{\pi}) - J(\pi) = \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t\geq 0}\gamma^t A^\pi(s_t, a_t)\right]. \tag{55}$$

Recognising the discounted state-visitation $d^{\tilde{\pi}}(s) = (1-\gamma)\sum_t \gamma^t P_{\tilde{\pi}}(s_t = s)$ converts the time sum into a state-action expectation:

$$J(\tilde{\pi}) - J(\pi) = \tfrac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^{\tilde{\pi}},\, a\sim\tilde{\pi}}[A^\pi(s,a)], \tag{56}$$

which is the state-distribution form quoted as (28).

For each prompt x, the inner optimisation in (42) reduces to

$$\max_{\pi(\cdot|x)} \sum_y \pi(y|x)\, r_\phi(x,y) - \beta \sum_y \pi(y|x)\log\tfrac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \tag{57}$$

subject to $\sum_y \pi(y|x) = 1$ and $\pi(y|x) \geq 0$. Introduce a Lagrange multiplier λ for the equality constraint; the resulting interior solution turns out strictly positive, so the non-negativity multipliers vanish at the optimum and may be omitted. Differentiating the Lagrangian and setting the result to zero,

$$r_\phi(x,y) - \beta\left[\log\tfrac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + 1\right] + \lambda = 0. \tag{58}$$

Rearranging,

$$\pi(y|x) = \pi_{\text{ref}}(y|x)\,\exp\!\left(\tfrac{r_\phi(x,y)}{\beta}\right)\exp\!\left(\tfrac{\lambda-\beta}{\beta}\right). \tag{59}$$

The y-independent factor $\exp((\lambda-\beta)/\beta)$ is pinned by the normalisation constraint:

$$\exp\!\left(\tfrac{\lambda-\beta}{\beta}\right) = \tfrac{1}{Z(x)}, \quad Z(x) = \sum_y \pi_{\text{ref}}(y|x)e^{r_\phi(x,y)/\beta}. \tag{60}$$

Substituting (60) into (59) recovers (43). Strict positivity of $\pi^\star$ is automatic from the exponential form.

Write $D(\theta) := \mathbb{E}_{s\sim d^\pi}[D_{\text{KL}}(\pi_{\theta_0}(\cdot|s)\,\|\,\pi_\theta(\cdot|s))]$ as a function of θ at fixed θ_0. Differentiating once,

$$\nabla_\theta D(\theta) = -\mathbb{E}_{s,\, a\sim\pi_{\theta_0}}[\nabla_\theta \log \pi_\theta(a|s)]. \tag{61}$$

At $\theta = \theta_0$ this expectation is over the same density we differentiate, so the score-mean-zero identity (13) gives $\nabla_\theta D(\theta)|_{\theta_0} = 0$.

For the second derivative, the supporting identity is

$$\mathbb{E}_{a\sim\pi_\theta}[\nabla_\theta^2 \log \pi_\theta(a|s)] = -F(\theta), \tag{62}$$

which follows by applying the quotient rule to $\nabla \log \pi_\theta = \nabla\pi_\theta/\pi_\theta$,

$$\nabla_\theta^2 \log \pi_\theta = \tfrac{\nabla_\theta^2 \pi_\theta}{\pi_\theta} - \nabla_\theta \log \pi_\theta\, \nabla_\theta \log \pi_\theta^\top, \tag{63}$$

then taking expectation: the first term integrates to $\nabla_\theta^2 \int \pi_\theta\, da = \nabla_\theta^2 1 = 0$ under the standard regularity conditions of Section II-B, leaving $-F(\theta)$. Combining (62) with (61) gives

$$\nabla_\theta^2 D(\theta)|_{\theta_0} = F(\theta_0), \tag{64}$$

which is the assertion that the Fisher matrix is the Hessian of the expected KL at the base point. The second-order Taylor expansion (33) follows directly: zero value, zero gradient, Hessian equal to F.

Define the per-example implicit reward $\hat{r}_\theta(x,y) := \beta\log\tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ and the preference margin $\Delta_\theta := \hat{r}_\theta(x,y^w) - \hat{r}_\theta(x,y^l)$, so that (45) reads $\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{\mathcal{D}}[\log\sigma(\Delta_\theta)]$. Using $\nabla \log\sigma(z) = \sigma(-z)\nabla z$,

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\mathbb{E}_{\mathcal{D}}[\sigma(-\Delta_\theta)\,\nabla_\theta\Delta_\theta]. \tag{65}$$

Since $\nabla_\theta \hat{r}_\theta(x,y) = \beta\nabla_\theta \log\pi_\theta(y|x)$, expanding $\nabla_\theta\Delta_\theta$ gives

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\beta\,\mathbb{E}_{\mathcal{D}}\Big[w_\theta \cdot \big(\nabla_\theta \log\pi_\theta(y^w|x) - \nabla_\theta \log\pi_\theta(y^l|x)\big)\Big], \tag{66}$$

with per-example weight

$$w_\theta(x,y^w,y^l) = \sigma\big(\hat{r}_\theta(x,y^l) - \hat{r}_\theta(x,y^w)\big). \tag{67}$$

Equation (67) sits near one when the model assigns higher implicit reward to the loser than the winner (a misclassified preference), and near zero when the winner is far ahead. DPO consequently performs implicit hard-example mining: examples already classified correctly contribute negligibly to the update, while ambiguous and misclassified examples dominate. As training progresses the gradient mass shifts onto whatever remains hard.

The same expression illuminates likelihood displacement. The update direction in (66) cares only about the difference between the score at y^w and at y^l. Provided this difference moves in the right direction, the loss decreases, and there is no penalty in the objective for the absolute log-likelihood $\log\pi_\theta(y^w|x)$ falling. If the parameter geometry makes it easier to lower $\log\pi_\theta(y^l|x)$ steeply than to raise $\log\pi_\theta(y^w|x)$, the optimiser will exploit that asymmetry, depressing the winner's likelihood alongside the loser's. The squared-loss alternative of IPO (46) eliminates this freedom by pinning the target log-ratio difference at a finite value instead of driving it to infinity.